

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 2 HOURS

SUMMER SEMESTER, 2024–2025
FULL MARKS: 120

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Simplify the following expression using the cut-and-try method and show that:

8 + 4
(CO2)
(PO1)

i) $F_1(a, b, c, d) = a + \bar{a}b + \bar{a}\bar{b}c + \bar{a}\bar{b}\bar{c}d = a + b + c + d$

ii) $F_2(X, Y, Z) = XY + \bar{X}Z + YZ = XY + \bar{X}Z$

Solution:

i) $F_1(a, b, c, d) = a + \bar{a}b + \bar{a}\bar{b}c + \bar{a}\bar{b}\bar{c}d = a + b + c + d$

$$F_1 = a + \bar{a}b + \bar{a}\bar{b}c + \bar{a}\bar{b}\bar{c}d$$

$$= (a + \bar{a}b) + \bar{a}\bar{b}c + \bar{a}\bar{b}\bar{c}d \quad (\text{associative law})$$

$$= (a + b) + \bar{a}\bar{b}c + \bar{a}\bar{b}\bar{c}d \quad (\text{since } a + \bar{a}b = a + b)$$

$$= (a + b) + \overline{(\bar{a} + \bar{b})}c + \bar{a}\bar{b}\bar{c}d \quad (\text{De Morgan: } \bar{a}\bar{b} = \overline{(\bar{a} + \bar{b})})$$

$$= (a + b + c) + \bar{a}\bar{b}\bar{c}d \quad (\text{using } X + \bar{X}Y = X + Y, X = a + b, Y = c)$$

$$= (a + b + c) + \overline{(\bar{a} + \bar{b} + \bar{c})}d \quad (\text{De Morgan: } \bar{a}\bar{b}\bar{c} = \overline{(\bar{a} + \bar{b} + \bar{c})})$$

$$= \boxed{a + b + c + d} \quad (\text{using } X + \bar{X}Y = X + Y, X = a + b + c, Y = d)$$

ii) $F_2(X, Y, Z) = XY + \bar{X}Z + YZ = XY + \bar{X}Z$

$$F_2 = XY + \bar{X}Z + YZ$$

$$= XY + \bar{X}Z + YZ(X + \bar{X}) \quad (\text{since } X + \bar{X} = 1)$$

$$= XY + \bar{X}Z + XYZ + \bar{X}YZ \quad (\text{distributive law})$$

$$= (XY + XYZ) + (\bar{X}Z + \bar{X}YZ) \quad (\text{associative and commutative laws})$$

$$= XY(1 + Z) + \bar{X}Z(1 + Y) \quad (\text{factorization})$$

$$= XY + \bar{X}Z \quad (\text{since } 1 + Z = 1, 1 + Y = 1)$$

$$= \boxed{XY + \bar{X}Z}$$

Rubrics:

• **Question 1 (8 Marks)**

- Correct use of Boolean algebra laws and identities. [4]
- Valid and logically consistent simplification steps. [3]
- Correct final simplified expression. [1]

• **Question 2 (4 Marks)**

- Correct use of Boolean algebra laws and identities. [2]
- Valid and logically consistent simplification steps. [1]
- Correct final simplified expression. [1]

- b) Simplify the following Boolean expression using the QuineMcCluskey (tabular) method. During the process, identify the prime implicants and essential prime implicants. Justify your choice of essential prime implicants in the final answer.

18
(CO3)
(PO2)

$$F(A, B, C, D) = \prod M(3, 4, 11, 12, 13, 15), d(7, 6)$$

Solution:

$$F(A, B, C, D) = \prod M(3, 4, 11, 12, 13, 15), d(6, 7)$$

Convert to sum of minterms (including don't cares as available):

$$F = \sum m(0, 1, 2, 5, 8, 9, 10, 14), d(6, 7)$$

Step 0: List Minterms by Number of 1's

Group	Minterm	Binary (ABCD)			
0	m_0	0	0	0	0
1	m_1	0	0	0	1
1	m_2	0	0	1	0
1	m_8	1	0	0	0
2	m_5	0	1	0	1
2	m_6^*	0	1	1	0
2	m_9	1	0	0	1
2	m_{10}	1	0	1	0
3	m_7^*	0	1	1	1
3	m_{14}	1	1	1	0

* don'tcare

Step 1: Mark All Minterms Used in Pairings

Group	Minterm	Binary				
0	m_0	0	0	0	0	✓
1	m_1	0	0	0	1	✓
1	m_2	0	0	1	0	✓
1	m_8	1	0	0	0	✓
2	m_5	0	1	0	1	✓
2	m_6^*	0	1	1	0	✓
2	m_9	1	0	0	1	✓
2	m_{10}	1	0	1	0	✓
3	m_7^*	0	1	1	1	✓
3	m_{14}	1	1	1	0	✓

* don'tcare

Step 2: Form Size-2 Implicants

Group	Matched Pairs	Binary				
0	m_0, m_1	0	0	0	—	✓
0	m_0, m_2	0	0	—	0	✓
0	m_0, m_8	—	0	0	0	✓
1	m_1, m_5	0	—	0	1	×
1	m_1, m_9	—	0	0	1	✓
1	m_2, m_6	0	—	1	0	✓
1	m_2, m_{10}	—	0	1	0	✓
1	m_8, m_9	1	0	0	—	✓
1	m_8, m_{10}	1	0	—	0	✓
2	m_5, m_7	0	1	—	1	×
2	m_6, m_7	0	1	1	—	×
2	m_6, m_{14}	—	1	1	0	✓
2	m_{10}, m_{14}	1	—	1	0	✓

✓ = combined further; × = cannot be combined prime implicant candidate

Step 3: Prime Implicants from Size-4 Combinations

Group	Matched Pairs	Binary				PI
0	m_0, m_1, m_8, m_9	—	0	0	—	$\overline{B}\overline{C}$
0	m_0, m_2, m_8, m_{10}	—	0	—	0	$\overline{B}\overline{D}$
1	m_2, m_6, m_{10}, m_{14}	—	—	1	0	$\overline{C}\overline{D}$
1	m_1, m_5	0	—	0	1	$\overline{A}\overline{C}\overline{D}$
2	m_5, m_7	0	1	—	1	$\overline{A}\overline{B}\overline{D}$
2	m_6, m_7	0	1	1	—	$\overline{A}\overline{B}\overline{C}$

The first three are size 4; the last three are size 2 that could not be expanded.

Step 4: Prime Implicant Coverage Table

PI	Minterms	0	1	2	5	8	9	10	14	
$\overline{B}\overline{C}$	0,1,8,9	×	×			×	⊗			✓
$\overline{B}\overline{D}$	0,2,8,10	×		×		×		×		
$\overline{C}\overline{D}$	2,6,10,14			×				×	⊗	✓
$\overline{A}\overline{C}\overline{D}$	1,5		×		×					
$\overline{A}\overline{B}\overline{D}$	5,7				×					✓
$\overline{A}\overline{B}\overline{C}$	6,7									

⊗ = essential prime implicant (sole cover of that minterm).

Selection of Essential Prime Implicants

Minterm 9 is covered only by $\overline{B}\overline{C}$; minterm 14 is covered only by $\overline{C}\overline{D}$.

These two essential PIs cover minterms {0, 1, 2, 8, 9, 10, 14}.

Remaining uncovered minterm: 5.

The PI $\overline{A}\overline{B}\overline{D}$ covers 5 (along with don't care 7) and PI $\overline{A}\overline{C}\overline{D}$ also covers 5 (along with

minterm 1) .Since both are of same size (2), **either** can be included in the solution (but not both).

Thus the minimal sum of products is:

$$F = \overline{B}\overline{C} + C\overline{D} + \overline{A}BD$$

or

$$F = \overline{B}\overline{C} + C\overline{D} + \overline{A}\overline{C}D$$

Rubrics:

- Correct conversion of the given function into an appropriate minterm representation and initial grouping of minterms (including don't-care terms where applicable). [2]
- Correct identification and formation of first-level implicant combinations. [2]
- Correct generation of higher-level implicants through successive combinations, with proper handling of don't-care terms. [3+3]
- Correct identification and listing of all prime implicants. [1]
- Construction of a valid prime implicant chart and identification of essential prime implicants. [3]
- Appropriate selection of additional prime implicants to obtain a minimal cover and justification of the final choice. [3]
- Correct final minimized Boolean expression, clearly presented. [1]

2. a) A combinational circuit converts the 4-bit binary representation of a **valid decimal digit** into its 4-bit 2's-complement representation. 4 + 4 ×
4 + 5
(CO5)
(PO3)
- i) Construct the complete truth table, including invalid BCD inputs and treat the outputs corresponding to invalid inputs as don't-care conditions. (The conversion of each binary input to its 4-bit two's complement representation need not be shown.)

Solution:

Inputs: A, B, C, D (A=MSB, weight 8).

Outputs: W, X, Y, Z (W=MSB).

The circuit computes the 2's complement of the input digit, i.e. 16 – input.

Invalid BCD inputs (10–15) produce don't-care outputs.

Input (BCD)				Output (2's complement)			
A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1
0	0	1	0	1	1	1	0
0	0	1	1	1	1	0	1
0	1	0	0	1	1	0	0
0	1	0	1	1	0	1	1
0	1	1	0	1	0	1	0
0	1	1	1	1	0	0	1
1	0	0	0	1	0	0	0
1	0	0	1	0	1	1	1
1	0	1	0	×	×	×	×
1	0	1	1	×	×	×	×
1	1	0	0	×	×	×	×
1	1	0	1	×	×	×	×
1	1	1	0	×	×	×	×
1	1	1	1	×	×	×	×

Rubrics:

- Correct identification and listing of all 16 possible 4-bit input combinations. [1]
- Correct output entries for valid BCD inputs (00001001) according to the required 4-bit two's-complement conversion. [2]
- Correct treatment and notation of invalid BCD inputs (10101111) as don't-care conditions. [1]

ii) Using Karnaugh maps, obtain the minimized Boolean expression for each output bit.

Solution:

Rubrics:

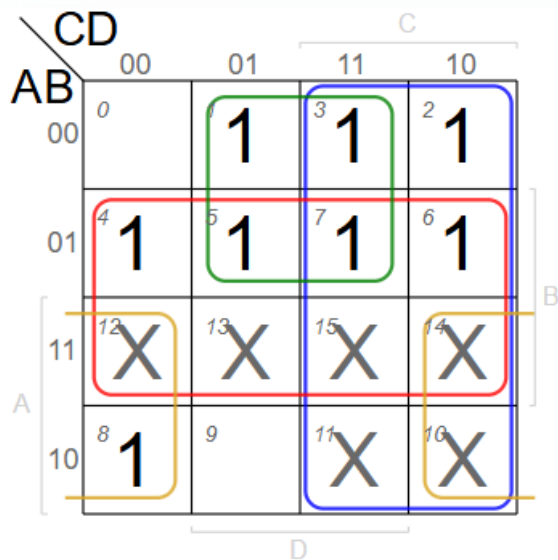
- Correct construction of K-maps for all 4 output variables with proper placement of 1s and don't-care conditions. [4 × 1]
- Correct grouping of minterms in each K-map using valid K-map rules (optimal or near-optimal groupings). [4 × 2]
- Correct minimization process leading to simplified Boolean expressions for all outputs. [4 × 1]
- Correct final simplified expressions for all 4 output functions. [4]

iii) Draw the logic implementation of the simplified expression of each output variable using only **NAND gates**.

Solution:

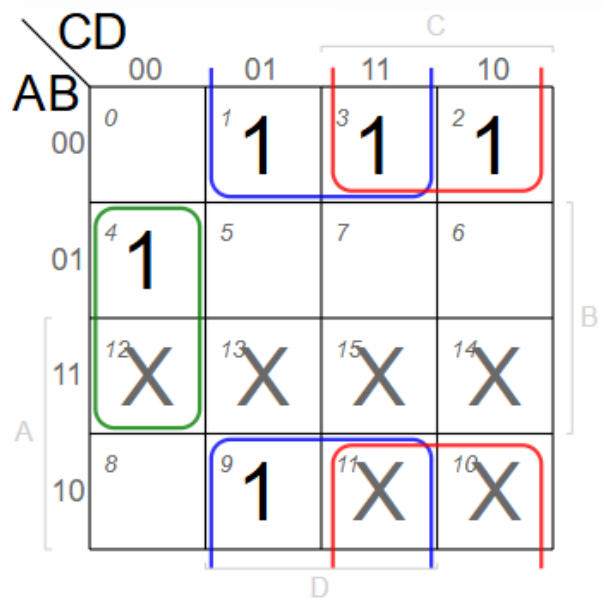
The proteus file can be found here [Link](#). The circuit diagram is in Figure 2.

Rubrics:



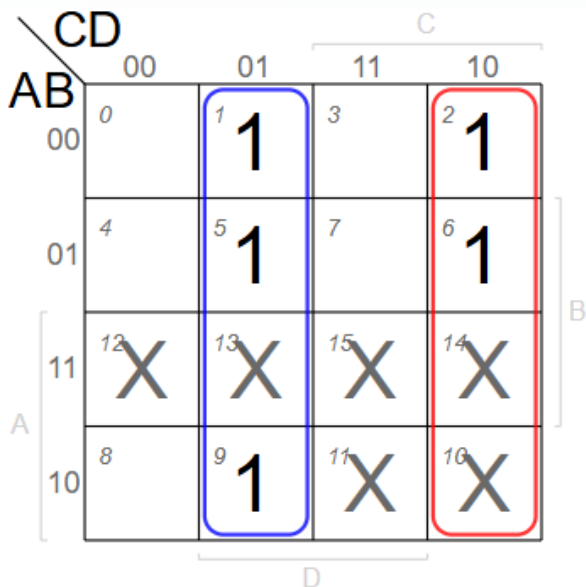
$$F = C + B + A'D + AD'$$

(a) W



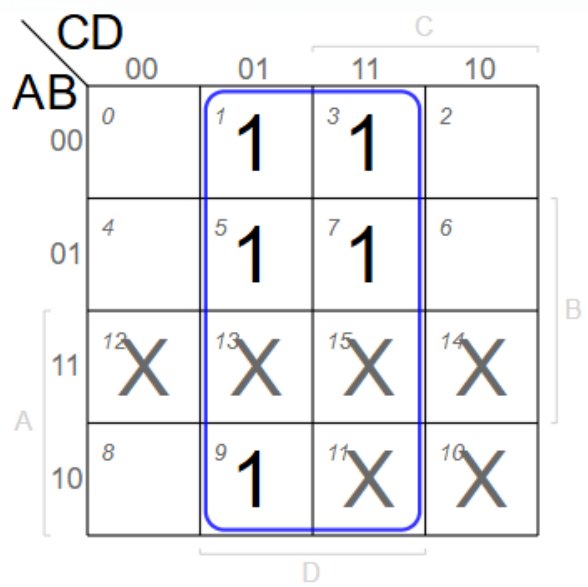
$$F = B'D + B'C + BC'D'$$

(b) X



$$F = C'D + CD'$$

(c) Y



$$F = D$$

(d) Z

Figure 1: Karnaugh maps for outputs W, X, Y, and Z

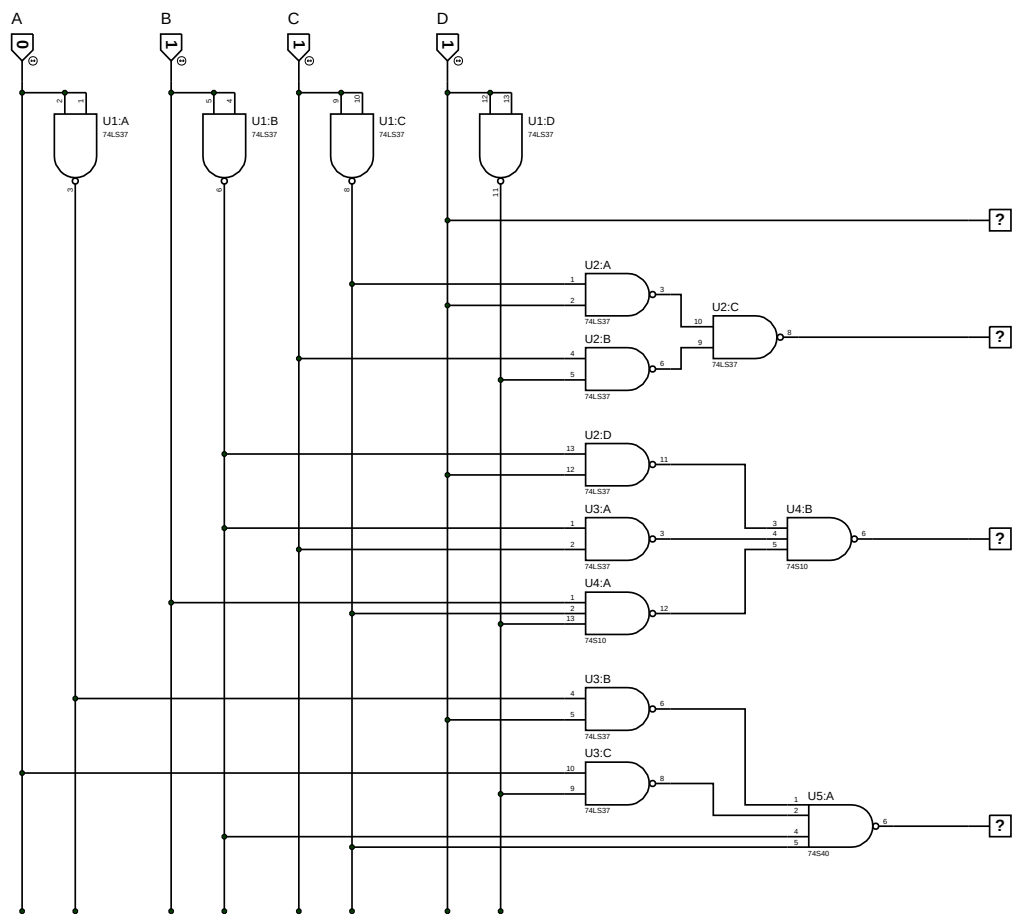


Figure 2: BCD to 2's Complement

- For correctly drawing the circuit for Z using NAND. [0.5]
- For correctly drawing the circuit for W and Y using NAND. [1.25×2]
- For correctly drawing the circuit for X using NAND. [2×1]

b) Convert $(10010110)_{\text{GRAY}}$ into its corresponding binary representation and determine whether the resulting binary number has even parity.

5
(CO1)
(PO1)

Solution:

Given: Gray code $(10010110)_{\text{GRAY}}$

Gray code to Binary

Let the Gray code be:

$$G = g_1g_2g_3g_4g_5g_6g_7g_8 = 10010110$$

The binary bits $B = b_1b_2b_3b_4b_5b_6b_7b_8$ are found using:

$$b_1 = g_1$$

$$b_i = b_{i-1} \oplus g_i \quad \text{for } i \geq 2$$

Now compute step by step:

$$b_1 = 1$$

$$b_2 = 1 \oplus 0 = 1$$

$$b_3 = 1 \oplus 0 = 1$$

$$b_4 = 1 \oplus 1 = 0$$

$$b_5 = 0 \oplus 0 = 0$$

$$b_6 = 0 \oplus 1 = 1$$

$$b_7 = 1 \oplus 1 = 0$$

$$b_8 = 0 \oplus 0 = 0$$

Thus, the binary number is: $(11100100)_2$

Check Parity

Count number of 1s in 11100100:

$$\text{Number of 1s} = 4$$

Since 4 is even, the binary number has **even parity**.

$$\text{Binary} = (11100100)_2, \quad \text{Even Parity}$$

Rubrics:

- For correctly converting each Gray code bit to its corresponding binary bit. [0.5×8]
- For correctly determining whether the resulting binary number has even parity. [1]

3. a) Convert the following decimal numbers into their corresponding BCD (Binary-Coded Decimal) representations. Then perform BCD addition to find the sum of the numbers.

$$(2856)_{10}, (7639)_{10}, (4562)_{10}$$

10
(CO2)
(PO1)

Solution:

Step 1: Convert each number to BCD

$$2856_{10} \rightarrow (0010\ 1000\ 0101\ 0110)_{BCD}$$

$$7639_{10} \rightarrow (0111\ 0110\ 0011\ 1001)_{BCD}$$

$$4562_{10} \rightarrow (0100\ 0101\ 0110\ 0010)_{BCD}$$

Step 2: Add $(2856)_{10}$ and $(7639)_{10}$ in BCD

Add digit by digit from LSD (units) to MSD (thousands).

If a nibble sum exceeds 9, add 6 (0110_2) to correct it.

$(2856)_{10}$	0010	1000	0101	0110
$(7639)_{10}$	0111	0110	0011	1001
Uncorrected sum	1001 ⁺¹	1110	1000 ⁺¹	1111
Correction	0110	0110		0110
Corrected sum	1 0000	0100	1001	0101

$$\text{First sum} = 1\ 0000\ 0100\ 1001\ 0101_{BCD} = 10495_{10}.$$

Step 3: Add $(10495)_{BCD}$ and $(4562)_{BCD}$ in BCD

Align digits: 10495 (5 digits) with 4562 (4 digits).

$(10495)_{10}$	0001	0000	0100	1001	0101
$(4562)_{10}$		0100	0101	0110	0010
Uncorrected sum	0001	0100 ⁺¹	1001 ⁺¹	1111	0111
Correction	---	---	0110	0110	
Corrected sum	0001	0101	0000	0101	0111

$$\text{Final sum} = 0001\ 0101\ 0000\ 0101\ 0111_{BCD}.$$

$$2856_{10} + 7639_{10} + 4562_{10} = 15057_{10}$$

Rubrics:

- For correctly converting the given decimal numbers into their corresponding BCD representations. [3]
- For correctly performing the first BCD addition. [2]
- For correctly performing the second BCD addition. [2]
- For correctly applying BCD correction in the first addition. [1]
- For correctly applying BCD correction in the second addition. [1]
- For obtaining the correct final BCD sum. [1]

- b) Perform the subtraction $(43307)_8 - (20431)_{10}$ by first converting both numbers into **base 12 number system**. Then carry out the subtraction using the **radix complement method** in **base 12**. You can skip the calculation of conversion of numbers between known bases and directly use values obtained from a calculator.

10
(CO1)
(PO1)

Solution:

We need to compute

$$(43307)_8 - (20431)_{10}$$

by first converting both numbers into base 12, then using the 12's complement method.

Conversion to base 12

$$\begin{aligned}(43307)_8 &= 4 \cdot 8^4 + 3 \cdot 8^3 + 3 \cdot 8^2 + 0 \cdot 8 + 7 \\ &= 4 \cdot 4096 + 3 \cdot 512 + 3 \cdot 64 + 7 \\ &= 16384 + 1536 + 192 + 7 = 18119_{10}\end{aligned}$$

$$\begin{aligned}18119_{10} &\xrightarrow{\div 12} 18119 = 12 \cdot 1509 + 11 (B) \\ &\xrightarrow{\div 12} 1509 = 12 \cdot 125 + 9 \\ &\xrightarrow{\div 12} 125 = 12 \cdot 10 + 5 \\ &\xrightarrow{\div 12} 10 = 12 \cdot 0 + 10 (A) \\ \Rightarrow (18119)_{10} &= (A59B)_{12}\end{aligned}$$

$$\begin{aligned}(20431)_{10} &\xrightarrow{\div 12} 20431 = 12 \cdot 1702 + 7 \\ &\xrightarrow{\div 12} 1702 = 12 \cdot 141 + 10 (A) \\ &\xrightarrow{\div 12} 141 = 12 \cdot 11 + 9 \\ &\xrightarrow{\div 12} 11 = 12 \cdot 0 + 11 (B) \\ \Rightarrow (20431)_{10} &= (B9A7)_{12}\end{aligned}$$

Add a leading 0 as sign bit

Both numbers are positive, so we write them with a 0 in the most significant (sign) position:

$$\text{Minuend: } (0A59B)_{12} \quad \text{Subtrahend: } (0B9A7)_{12}$$

12's complement of the subtrahend (5 digits)

$$12\text{'s complement} = 12^5 - (0B9A7)_{12}$$

Digitwise (11's complement +1):

$$11 - 0 = B, \quad 11 - B = 0, \quad 11 - 9 = 2, \quad 11 - A = 1, \quad 11 - 7 = 4$$

$$\Rightarrow (B0214)_{12} \xrightarrow{+1} (B0215)_{12}$$

Add minuend and the 12's complement

$$\begin{array}{rcccll}
 & & & +1 & & \\
 & 0 & A & 5 & 9 & B \\
 + & B & 0 & 2 & 1 & 5 \\
 \hline
 & B & A & 7 & B & 4
 \end{array}$$

The difference is **negative** since sign bit is **B**, so, to get the original value we need to convert it to 12's complement.

$$(BA7B4)_{12} \xrightarrow{\text{11's Complement}} (01407)_{12} \xrightarrow{+1} (01408)_{12}$$

So the difference using 12's complement is

$$-(01408)_{12}$$

Rubrics:

- For correctly converting both given numbers into their corresponding base-12 representations. [2]
- For correctly obtaining the 12's complement of the subtrahend (including the 11's complement and adding 1). [2]
- For correctly performing the base-12 addition. [3]
- For correctly identifying the sign of the result and taking the 12's complement of the obtained sum to determine the magnitude. [2]
- For writing the correct final answer in base 12. [1]

c) The following calculation is performed in a number system with base x :

$$(15)_x + (4)_x = (22)_x$$

Find the value of the base x .

10
(CO1)
(PO1)

Solution:

$$(15)_x + (4)_x = (22)_x$$

Convert each number from base x to base 10:

$$(15)_x = 1 \cdot x + 5 = x + 5,$$

$$(4)_x = 4,$$

$$(22)_x = 2 \cdot x + 2 = 2x + 2.$$

Substitute into the given equation:

$$(x + 5) + 4 = 2x + 2.$$

Simplify:

$$x + 9 = 2x + 2,$$

$$9 = x + 2,$$

$$x = 7.$$

Since the largest digit used is 5, the base must satisfy $x > 5$, and $x = 7$ is valid.

Answer : 7

Rubrics:

- For correctly converting $(15)_x$, $(4)_x$, and $(22)_x$ into their decimal expressions. [3]
- For correctly forming the equation $(x + 5) + 4 = 2x + 2$. [3]
- For correctly solving the equation to obtain $x = 7$. [3]
- For verifying that $x = 7$ is a valid base (i.e., $x > 5$). [1]

4. a) Draw a 3-bit Adder-Subtractor circuit using Full Adders (FA). You may represent the Full Adders as block diagrams and do not need to show their internal gate-level implementation. Illustrate the operation of the circuit using two numerical examples, one demonstrating addition and the other demonstrating subtraction.

4 + 6
(CO3)
(PO2)

Solution:

The solution is similar to that given in L5_Combinational Circuits.pdf Section 2.5
Any valid case will be accepted.

Rubrics:

- For correctly drawing the 3-bit Adder-Subtractor circuit using Full Adder (FA) blocks, including the XOR gates for mode control and proper carry connections. [4]
- For correctly illustrating the addition operation with a suitable numerical example, including the input values and the correct output. [3]
- For correctly illustrating the subtraction operation with a suitable numerical example, including the use of 2's complement, the input values, and the correct output. [3]

- b) You are required to design a combinational circuit that takes two 4-bit binary numbers A and B as inputs and produces the result of the subtraction $A - B$ using the **diminished radix complement method**. The following conditions must be considered while designing the circuit:

12 + 3
(CO5)
(PO3)

- B is always subtracted from A , but B is not necessarily smaller than A .
- You may use block diagrams of standard components such as a 4-bit adder (which accepts two 4-bit inputs, a carry-in, and produces a 4-bit sum and a carry-out), along with any required logic gates.
- The output must consist of a single 4-bit value representing the magnitude of the difference, along with one additional bit indicating the sign of the result (positive or negative). Separate outputs for positive and negative results are not allowed.

Also, add a short description of the logic behind the construction of the circuit.

Solution:

Try to figure it out by yourself.

Rubrics:

- For correctly generating the diminished radix complement (1's complement) of the subtrahend using logic gates. [2]
- For correctly connecting the 4-bit adder with the minuend, complemented subtrahend, and appropriate carry-in. [3]
- For correctly detecting whether the result is positive or negative using the carry-out. [2]
- For correctly obtaining the magnitude of the result by conditionally complementing the adder output when the result is negative. [3]
- For correctly showing the final 4-bit magnitude output and the separate sign bit. [2]
- For clearly explaining the logic behind the circuit construction and its operation for both positive and negative results. [3]

- c) Is the circuit shown in Figure 3 a combinational or a sequential circuit? Justify your answer. Also, suggest appropriate modifications to convert it into the other type of circuit.

3 + 2
(CO3)
(PO2)

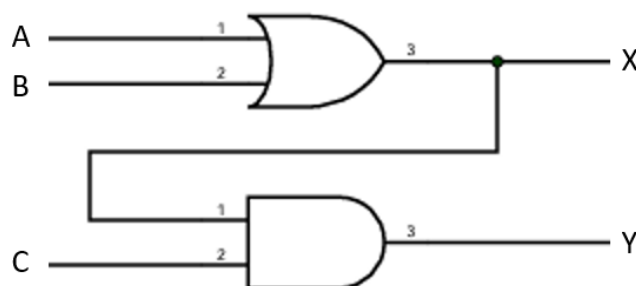


Figure 3: Circuit diagram for Question 4.c

Solution:

The circuit in Figure 3 is a **combinational circuit** because its outputs depend only on the **current input values**. It does not contain any memory elements or feedback paths, so the outputs are independent of any previous input or memory states or maybe something simpler.

For the second part there are multiple answers, one of them is if X was feeded into B.

Rubrics:

- For correctly identifying the given circuit as a combinational circuit. [1]
- For correctly justifying the answer by explaining that the outputs depend only on the present inputs and that the circuit contains no memory elements or feedback. [2]
- For correctly suggesting suitable modifications to convert the circuit into a sequen-

tial circuit (e.g., introducing flip-flops/latches and a clock signal to provide memory). [2]

MARKS SUMMARY TABLE

Question	Marks	Keywords
1.a	12	Simplification using cut and try
1.b	18	Tabular Method
2.a	25	Kmaps
2.b	5	Gray Code
3.a	10	BCD Addition
3.b	10	Subtraction using Complements
3.c	10	Unknown Base
4.a	10	Adder Subtractor Circuit
4.b	15	2's Complement Circuit
4.c	5	Combinational vs Sequential
Total	120	